

Session :-1

Question :- 1. Name 5 IT job roles and write one line about what each role does.

Answer

- :- Software Developer – Writes code to build software and applications.
- :- Software Tester (QA) – Tests software to find bugs and ensure quality.
- :-UI/UX Designer – Designs user-friendly and attractive application interfaces.
- :- Data Analyst – Analyzes data to help businesses make better decisions.
- :-AI Engineer – Develops Artificial Intelligence and Machine Learning systems.

Question :-2. List 3 product companies and 3 service companies with one example of work each does.

Answer :- Product Companies

- :- Google – Develops products like Google Search and Android.
- :- Zoho – Creates business software such as Zoho CRM.
- :- Microsoft – Develops Windows, Microsoft Office, and Azure.

Service companies

- :- TCS (Tata Consultancy Services) – Provides software development and IT consulting.
- :- Infosys – Offers digital transformation and software services.
- :- Wipro – Provides cloud, cybersecurity, and IT support services.

Question :- 3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Answer

- :- GitHub – No official full form – A platform for storing, managing, and sharing source code using Git.
- :- Slack – No official full form – A team communication and collaboration tool.
- :- Jira – No official full form – A project management and bug-tracking tool.
- :- IDE – Integrated Development Environment – Software used to write, run, and debug programs.
- :- API – Application Programming Interface – Allows different software applications to communicate with each other.

Question :- 4. Name any 5 programming languages and mention one popular use case for each.

Answer

- :- Python – AI, Machine Learning, and Web Development.
- :- Java – Android applications and Enterprise software.
- :- C – Operating systems and Embedded systems.
- :- JavaScript – Interactive websites and Web applications.
- :- C++ – Game development and High-performance software.

Question :- 5.What is the difference between a developer and a tester? (2 lines)

Answer

:- Developer – Writes code to create software and add new features.

:- Tester (QA) – Tests the software, finds bugs, and ensures it works correctly before release.

Session :-2

Question :-1. What are the three main layers of software architecture? Write one line about each.

Answer

:- UI (User Interface): The part of the application that users see and interact with.

:- Business Logic: The layer that processes data and performs the application's main functions.

:- Database: The layer that stores and manages the application's data.

Question :-2. Define client and server in your own words with one real-life example.

Answer

:- Client: A device or application that sends a request for information or a service.

:- Server: A computer that receives the request, processes it, and sends back the required data.

:- Example: When you open Instagram on your phone, your phone (client) requests your feed, and Instagram's server sends the latest posts back to your phone.

Question :-3. Write 2 differences each between desktop apps, web apps, and mobile app

Answer

:- Desktop Apps

1. Installed on a computer or laptop.

2. Can work offline in many cases.

:- Web Apps

1. Run in a web browser without installation.

2. Usually require an internet connection.

:- mobile Apps

1. Installed on a smartphone or tablet.

2. Can use phone features like the camera, GPS, and sensors.

Question :-4. Name any 3 databases and mention whether each is SQL or NoSQL.

Answer :- Database Type

:- SQL : MySQL, PostgreSQL

:- NoSQL : MongoDB

Session :-3

Question :-1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Answer

:-HTTP – HyperText Transfer Protocol: Used to transfer web pages between a browser and a web server.

:-HTTPS – HyperText Transfer Protocol Secure: A secure version of HTTP that encrypts data using SSL/TLS.

:-FTP – File Transfer Protocol: Used to upload and download files between computers over a network.

:-SMTP – Simple Mail Transfer Protocol: Used to send emails from one mail server to another.

Question :-2.What is the difference between IPv4 and IPv6? (2 lines)

Answer

:-IPv4 uses 32-bit addresses (e.g., 192.168.1.1) and supports about 4.3 billion unique addresses.

:-IPv6 uses 128-bit addresses (e.g., 2001:db8::1) and supports a much larger number of unique addresses.

Question :-3.Run nslookup google.com on your system and write down the IP address you get.

Answer

Example output:

:-IPv4: 142.250.183.14

:-Note: The IP address may be different on your system depending on your location and Google's servers.

Question :-4.What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Answer

:-200 OK – The request was successful.

:-301 Moved Permanently – The webpage has permanently moved to a new URL.

:-404 Not Found – The requested page was not found.

:-500 Internal Server Error – The server encountered an unexpected error.

:-403 Forbidden – Access to the requested resource is denied.

Question :-5.Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Answer

:-Root (.) – The top level of the DNS hierarchy (usually hidden).

:-TLD (Top-Level Domain) – .com

:-Domain – google

:-Subdomain – www

:-So, in www.google.com◆:

:-Root → .

:-TLD → .com

:-Domain → google

:-Subdomain → www

Session :-4

Question :-1. List all 6 phases of SDLC in order with one line about each.

Answer

:-Requirement Gathering – Collect and understand the customer's needs and project requirements.

:-Analysis – Analyze the requirements and plan the system, resources, and feasibility.

:-Designing – Create the software architecture, database, and user interface design.

:-Implementation (Development) – Write the code and build the software according to the design.

:-Testing – Test the software to find and fix bugs and ensure it works correctly.

:-Maintenance – Update, improve, and fix the software after it is deployed.

Question :-2. Write 3 differences between Agile and Waterfall models.

Answer

Agile.

:-Development is done in small iterations (Sprints).

:-Changes can be made during the project.

:-Customer feedback is taken regularly.

Waterfall.

:-Development follows one fixed sequence of phases.

:-Changes are difficult after development starts.

:-Customer feedback usually comes at the end of the project.

Question :-3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Answer

:-Product Owner (PO) – Decides project requirements and prioritizes the work.

:-Scrum Master – Helps the team follow Scrum practices and removes obstacles.

:-Development Team – Designs, develops, tests, and delivers the software.

Question :-4. What is a Sprint in Scrum? Mention its typical duration

Answer

:-A Sprint is a fixed time period during which the Scrum team completes a set of planned tasks and delivers a working feature.

:-Typical duration: 1–4 weeks (most commonly 2 weeks).

Question :-5. What do these Jira board columns mean?

Answer

:-Backlog – List of all tasks and user stories waiting to be worked on.

:-To Do – Tasks that are planned but have not been started yet.

:-In Progress – Tasks that are currently being worked

:-Done – Tasks that have been completed and finished.

Session :-5

Question :-1. write the Git commands

Answer

:- Initialize a repository —git init

:-Stage all files —git add.

:-Commit with a message —git commit-m initial commit

:-Push to remote —git push origin main

:-Pull latest changes —git pull origin main

Question :-2. Difference between git fetch and git pull (2 lines)

Answer

:-git fetch downloads the latest changes from the remote repository but does not merge them into your current branch.

:-git pull downloads the latest changes and automatically merges them into your current branch.

Question :-3. Create a GitHub repository and share the link

Answer

https://github.com/rathodvijay9832/04july_vijay_SE

Question :-4. Difference between a Fork and a Pull Request (2 lines)

Answer

:-A Fork is a personal copy of someone else's GitHub repository in your own account.

:-A Pull Request (PR) is a request to merge your changes into the original repository.

Question :-5. Create a branch, commit, and merge back to main

Answer

:-git checkout -b feature-test

:-echo "New feature" > feature.txt

:-git add feature.txt

:-git commit -m "Add feature.txt"

:-git checkout main

:-git merge feature-test